SONG JIAN

Tokyo, Japan

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https://jtrneo.github.io   ResearchGate   LinkedIn

Google Scholar   GitHub   ORCID

Research Interests

- Remote sensing, 3D vision, high-fidelity synthetic 3D environments, advanced scene generation, transfer learning, scalable 3D understanding, and geospatial artificial intelligence.

Education

The University of Tokyo — **Oct. 2021 – Jun. 2025**
Ph.D. in Engineering — *Chiba, Japan*
- Laboratory: Machine Learning and Statistical Data Analysis Lab
- Supervisor: Prof. Naoto Yokoya
- Research topic: Synthetic data-driven 3D semantic reconstruction in remote sensing

Waseda University — **Oct. 2019 – Mar. 2021**
Master of Engineering — *Fukuoka, Japan*
- Major: Information Engineering
- Research topic: Flow-guided video object detection

Wuhan University — **Sep. 2018 – Jun. 2021**
Master of Engineering — *Wuhan, China*
- Major: Software Engineering
- Research topic: Remote sensing image semantic segmentation

Fujian Agriculture and Forestry University — **Sep. 2011 – Jun. 2015**
Bachelor of Agriculture — *Fuzhou, China*
- Major: Forestry Information Engineering

Employment History

RIKEN Center for Advanced Intelligence Project — **Jul. 2025 – Present**
Postdoctoral Researcher, Geoinformatics Team — *Tokyo, Japan*
- Advisor: Prof. Naoto Yokoya
- Research focus: high-fidelity synthetic 3D environments, advanced scene generation, transfer learning, remote sensing image synthesis, and scalable global 3D understanding.

RIKEN Center for Advanced Intelligence Project — **Apr. 2021 – Jun. 2025**
Research Part-timer, Geoinformatics Team — *Tokyo, Japan*
- Supervisor: Prof. Naoto Yokoya

The University of Tokyo — **Oct. 2021 – May 2024**
Research Assistant — *Tokyo, Japan*
- Supervisor: Prof. Naoto Yokoya

Inception Institute of Artificial Intelligence — **Mar. 2019 – Aug. 2019**
Research Intern — *Abu Dhabi, UAE*
- Supervisor: Dr. Fan Zhu

Land Satellite Remote Sensing Application Center — **May 2018 – Sep. 2018**
Research Intern — *Beijing, China*
- Supervisor: Dr. Yuhang Gan

Fundings and Awards

JSPS Grant-in-Aid for Early-Career Scientists (KAKENHI 26K21262) **Apr. 2026 – Mar. 2031**
PI; 4,550,000 JPY *[Link]*
– Project: "Open Synthetic Data Engine for Physics-Based Geospatial Foundation Models"

RIKEN BAIHO Award **Mar. 2025**
Awarded for "Development of remote sensing image synthesis technology"

AY2024 AI Center Fusion Research Promotion Fund **Dec. 2023**
1,600,000 JPY

SPRING GX Self-directed and Integrated Project Research **Jul. 2024**
250,000 JPY

SPRING GX Fellowship **Apr. 2024**
180,000 JPY per month

AY2023 AI Center Fusion Research Promotion Fund **Dec. 2023**
2,000,000 JPY

AY2023 GSFS Challenging New Area Doctoral Research Grant **Mar. 2023**
600,000 JPY

AY2020 JEES–SoftBank AI Scholarship **Feb. 2020**
1,000,000 JPY

Kitakyushu Science and Research Park Scholarship **Sep. 2019**
300,000 JPY

Publications (∗ denotes equal contribution)

5.1 Preprints

- H. Chen∗, **J. Song**∗, W. Xuan∗, J. Wang∗, H. Qi, Z. Zhou, P. Dai, O. Dietrich, E. Gutierrez, L. Bromly, E. Nemni, Y. Ou, J. Zhao, Z. Zheng, Y. Xu, R. Hänsch, W. Jiao, M. Chini, C. Persello, J. Xia, S. Lu, L. Wang, Z. Zhu, E. Shelhamer, J. Chanussot, K. Schindler, and N. Yokoya, "Earth Observation for Disaster Mapping: Benchmarks, Methods, Challenges and Future Perspectives," *Preprint*, 2026. [Paper] [Code]

5.2 Selected Journal Articles

- H. Chen, C. Lan, **J. Song**, D. Ibañez, J. Xia, K. Schindler, and N. Yokoya, "Multimodal Remote Sensing Change Detection: An Image Matching Perspective," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 233, pp. 487–501, 2026. [Link]
- **J. Song**, H. Chen, and N. Yokoya, "Enhancing Monocular Height Estimation via Sparse LiDAR-Guided Correction," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 232, pp. 155–171, 2026. [Link]
- H. Chen, **J. Song**, O. Dietrich, C. Broni-Bediako, W. Xuan, J. Wang, X. Shao, Y. Wei, J. Xia, C. Lan, K. Schindler, and N. Yokoya, "Bright: A Globally Distributed Multimodal Building Damage Assessment Dataset With Very-High-Resolution for All-Weather Disaster Response," *Earth System Science Data*, vol. 17, no. 11, pp. 6217–6253, 2025. (IEEE GRSS Data Fusion Contest 2025 official dataset) [Link]
- J. Xia, H. Chen, C. Broni-Bediako, Y. Wei, **J. Song**, and N. Yokoya, "OpenEarthMap-SAR: A Benchmark Synthetic Aperture Radar Dataset for Global High-Resolution Land Cover Mapping," *IEEE Geoscience and Remote Sensing Magazine*, vol. 13, no. 4, pp. 476–487, 2025. [Link]
- H. Chen∗, **J. Song**∗, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote Sensing Change Detection With Spatiotemporal State Space Model," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–20, 2024. (ESI Hot Paper & ESI Highly Cited Paper) [Link]
- C. Broni-Bediako, J. Xia, **J. Song**, H. Chen, M. Siam, and N. Yokoya, "Generalized Few-Shot Semantic Segmentation in Remote Sensing: Challenge and Benchmark," *IEEE Geoscience and Remote Sensing Letters*, pp. 1–5, 2024. [Link]
- H. Chen, C. Lan, **J. Song**, C. Broni-Bediako, J. Xia, and N. Yokoya, "ObjFormer: Learning Land-Cover Changes From Paired OSM Data and Optical High-Resolution Imagery via Object-Guided Transformer," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–22, 2024. [Link]

- X. Shao, H. Chen, K. Magson, J. Wang, **J. Song**, J. Chen, and J. Sasaki, "Deep Learning for Multi-Label Classification of Coral Conditions in the Indo-Pacific via Underwater Photogrammetry," *Aquatic Conservation: Marine and Freshwater Ecosystems*, vol. 34, no. 9, e4241, 2024. (Front cover article) [Link]
- H. Chen, **J. Song**, C. Wu, B. Du, and N. Yokoya, "Exchange Means Change: An Unsupervised Single-Temporal Change Detection Framework Based on Intra- and Inter-Image Patch Exchange," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 206, pp. 87–105, 2023. [Link]

5.3 Selected Conference Papers

- J. Wang*, W. Xuan*, H. Qi, Z. Liu, K. Liu, Y. Wu, H. Chen, **J. Song**, J. Xia, Z. Zheng, and N. Yokoya, "DisasterM3: A Remote Sensing Vision-Language Dataset for Disaster Damage Assessment and Response," *Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2025. [Link]
- **J. Song**, H. Chen, W. Xuan, J. Xia, and N. Yokoya, "SynRS3D: A Synthetic Dataset for Global 3D Semantic Understanding from Monocular Remote Sensing Imagery," *Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2024. (Spotlight Paper, top 3.08%) [Paper] [Project] [Code]
- N. Yokoya, J. Xia, C. Broni-Bediako, **J. Song**, and H. Chen, "OpenEarthMap Benchmark Suite and Its Applications," *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2024. (Oral Presentation) [Link]
- H. Chen, **J. Song**, and N. Yokoya, "Change Detection Between Optical Remote Sensing Imagery and Map Data via Segment Anything Model (SAM)," *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2024. (Oral Presentation) [Link]
- **J. Song**, H. Chen, and N. Yokoya, "SyntheWorld: A Large-Scale Synthetic Dataset for Land Cover Mapping and Building Change Detection," *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024. [Link]
- **J. Song**, B. Adriano, and N. Yokoya, "Disaster Detection from SAR Images With Different Off-Nadir Angles Using Unsupervised Image Translation," *2nd CDCEO Workshop at IJCAI*, 2022. [Link]

Academic Services

- Reviewer: IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- Reviewer: IEEE Transactions on Image Processing (TIP)
- Reviewer: IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- Reviewer: Engineering Applications of Artificial Intelligence (EAAI)
- Reviewer: Conference on Neural Information Processing Systems (NeurIPS) 2025
- Reviewer: IEEE Transactions on Geoscience and Remote Sensing (TGRS)
- Reviewer: ISPRS Journal of Photogrammetry and Remote Sensing
- Reviewer: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- Reviewer: IEEE Access
- Co-organizer: IEEE GRSS Data Fusion Contest 2025 [Link]
- Co-organizer: OpenEarthMap Land Cover Mapping Few-Shot Challenge at L3D-IVU CVPR 2024 Workshop [Link]